

WEEK - 01

FCFS

```
#include <stdio.h>

int main() {
    int n, i, j;
    int at[10], bt[10], ct[10], wt[10], tat[10], p[10];
    float avg_wt = 0, avg_tat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        p[i] = i + 1; // Process IDs
        printf("Enter AT and BT for P%d: ", i + 1);
        scanf("%d%d", &at[i], &bt[i]);
    }

    // Sort processes based on Arrival Time (Essential for FCFS)
    for(i = 0; i < n - 1; i++) {
        for(j = 0; j < n - i - 1; j++) {
            if(at[j] > at[j+1]) {
                // Swap Arrival Time
                int temp = at[j];
                at[j] = at[j+1];
                at[j+1] = temp;
                // Swap Burst Time
                temp = bt[j];
                bt[j] = bt[j+1];
                bt[j+1] = temp;
                // Swap Process ID
                temp = p[j];
                p[j] = p[j+1];
                p[j+1] = temp;
            }
        }
    }
}
```

```

// Calculate Completion, TAT, and WT
int current_time = 0;
for(i = 0; i < n; i++) {
    // If CPU is idle (gap between processes)
    if(current_time < at[i]) {
        current_time = at[i];
    }

    current_time += bt[i];
    ct[i] = current_time;
    tat[i] = ct[i] - at[i];
    wt[i] = tat[i] - bt[i];

    avg_wt += wt[i];
    avg_tat += tat[i];
}

printf("\nP\tAT\tBT\tCT\tWT\tTAT\n");
for(i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\t%d\n", p[i], at[i], bt[i], ct[i], wt[i],
tat[i]);
}

printf("\nAverage Waiting Time = %.2f", avg_wt/n);
printf("\nAverage Turnaround Time = %.2f\n", avg_tat/n);

return 0;
}

```

OUTPUT

```
Enter number of processes: 3
Enter AT and BT for P1: 0 12
Enter AT and BT for P2: 3
4
Enter AT and BT for P3: 20
5

P      AT      BT      CT      WT      TAT
P1      0      12      12      0      12
P2      3       4      16      9      13
P3     20       6      26      0       6

Average Waiting Time = 3.00
Average Turnaround Time = 10.33

Process returned 0 (0x0)  execution time : 20.081 s
Press any key to continue.
```